Dell PowerMax File

Quick Start Guide

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Rev. 01

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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PREFACE

As part of an effort to improve its product lines, revisions of the software and hardware are periodically released. Therefore, some functions described in this document might not be supported by all versions of the software or hardware currently in use. The product release notes provide the most up-to-date information on product features.

Contact your Dell EMC technical support professional if a product does not function properly or does not function as described in this document.

NOTE: This document was accurate at publication time. Go to Dell Online Support (https://support.emc.com) to ensure that you are using the latest version of this document.

Where to get help

Dell support, product, and licensing information can be obtained as follows:

Product information	For product and feature documentation or release notes, go to the PowerMax Documentation page at : https://www.dell.com/support/kbdoc/en-us/000189115/powermax-info-hub-product-documentation-videos?lang=en.
Troubleshooting	For information about products, software updates, licensing, and service, go tohttps://www.dell.com/supportand locate the appropriate product support page.
Technical support	For technical support and service requests, go to https://www.dell.com/support and locate the Service Requests page. To open a service request, you must have a valid support agreement. Contact your Sales Representative for details about obtaining a valid support agreement or to answer any questions about your account.

Revision history

Document revision	Date	Description
01	July 2022	Initial release.

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1

Getting Started with Configuration

This chapter provides information that you should know prior to configuring PowerMax File on a storage array.

NOTE: For known limitations related to PowerMax File please refer to the Dell PowerMaxOS 10.0.0.0 Release Notes.

Topics:

- Prerequisites
- Terminologies
- Components used in a system with PowerMax File
- Configuration information

Prerequisites

NOTE: PowerMax File (File) relies on PowerMax Embedded Management guest for access to Unisphere for PowerMax. Ensure the management guest is fully functional before attempting to configure File.

Before starting the configuration process, complete the following tasks:

- Verify that your Dell support representative has configured the VMAX and PowerMax systems to call home with EMS Secure Remote Services (ESRS).
- Verify that MGMT and File applications are in healthy state using a container health check. Refer Serviceability guide.
- Gather and record configuration information using the Setup worksheets, as described in Configuration information.

Terminologies

Terms	Definition
File Cluster	A set of File nodes that operate together. A File cluster can support a maximum of 8 nodes. The first node by default operates as the Cluster controller and any node can take over Cluster controller activities in the event of first node failure.
NAS Server	A virtual container that hosts one or more Filesystems. The NAS Server has a unique name, one or more IP addresses, exposes NAS protocols such as NFS, SMB, FTP and SFTP. Naming services like DNS, LDAP, NIS are configured on a NAS Server. Users and groups may be defined locally with local files. NAS Server facilitates protection by remote NDMP software. PowerMax File supports 3-way NDMP. NAS Server also allows configuration for virus checker. A NAS Server runs by default on the primary node. In case of a node failure, the NAS Server is moved automatically to its backup node.
NAS Server primary node	One of the File cluster node on which the NAS Server currently run.
NAS Server backup node	One of the File cluster nodes on which the NAS Server is backed up for fault tolerance. The NAS Server will be moved to this node during any failover event.
File Node	Facilitates hosting one or more NAS Servers. Each File node exposes one or more physical Ethernet SLICs to enable access to Filesystem data.

Components used in a system with PowerMax File

The following components are used for File:

- Control Station (CS): Used by Dell Customer Support to configure File.
- Unisphere for PowerMax: Provides management functions to the file components.
- NAS Servers: Clients communicate with a NAS Server using NFS or SMB or both protocols. Clients are physically connected to the NAS Server through I/O modules on the storage array that are assigned to the NAS Server.

Configuration information

Before starting the configuration process, it is recommended to complete the following worksheets. Since this information will be required later while running the Initial Configuration Wizard for File, the worksheets are organized by wizard page:

1. Set up Network worksheet
2. Setup NAS Server worksheet
3. Create Filesystem worksheet
4. Create a SMB share worksheet

Before you begin, ensure you have the following identified/configured.

Is the NTP configured in Unisphere for PowerMax? NTP is required for NFS Kerberos and SMB operations. It can be configured from Unisphere for PowerMax Serviceability section.	Yes/No
How many subnets exists in Customer environment that will be part of File configuration? Note: PowerMax File can support heterogeneous network environment. Identifying subnets helps File to optimize network configuration like failover, bonding. Refer Networking section in Chapter 3 for more details.	

Setup Network worksheet

File Node (1-8)	SLIC Slot	LIC Port	IPv4 Subnet/Prefix Length (Like, 192.10.5.0/21)	IPv6/Prefix Length	NIC Bonding required? (Only within same node)	NIC Bonding Type (Active-Backup / DLA)
<1>	<1>	<4>	<10.243.5.0/21>	<2001:db8:1234:002A::0/64>	<Yes with SLIC Port 5>	<DLA>

Setup NAS Server worksheet

NAS Server Name	Primary Node	Backup Node	Subnets	IP Address (V4/V6)	Gateway	Role (Production/ Backup)

NAS Server Name	Primary Node	Backup Node	Subnets	IP Address (V4/V6)	Gateway	Role (Production/ Backup)

DNS Domain Name 1	
DNS Servers for that domain	
DNS Domain Name 2	
DNS Servers for that domain	
Protocol (checkbox)	<------>UDP < > TCP
NIS Domain Name	
NIS Servers	

Filesystem worksheet

Filesystem Name	
Hosting NAS Server	
Filesystem Size (in mb/gb/tb)	
Compression/Dedupe Enabled?	
Traditional / Transactional (VMWare) Filesystem?	
If Transactional, block size?	
FLR?	FLR-C/FLR- E?
High Water Mark (ranges from 50-99; defaultsto 90)	
Hard limit for user storage(in mb/gb/tb)	
Soft limit for user storage(in mb/gb/tb)	
Hard limit for user directories (in mb/gb/tb)	
Soft limit for user directories (in mb/gb/tb)	
Enforce Hard limits?	
Grace period for Storage (in days)	
Grace period for File (in days)	

Create a SMB share worksheet

NAS Server name	
SMB Type	Standalone < --- > Active Directory < -- >

SMB Configuration	
Windows Domain	
Computer Name	
NetBIOS Name (Optional)	
Administrator password (obtain from customer)	
Join the domain?	Yes / No?
Domain Admin Username(Windows)	
Domain Admin Password(Windows)	
OrganizationalUnit (Windows)	
NetBIOS Name	
SMB Share Name	
SMB Share path name	
Comment (Optional description)	
User Limit (Optional)	

2

Setup PowerMax File Configuration

Topics:

- Logging into Dell Unisphere for PowerMax
- Walk with Initial Configuration Wizard
- Launching the Initial Configuration Wizard (ICW)
- ICW Workflow – Configure Subnets
- ICW Workflow – Create NAS Server
- ICW Workflow – Create File System
- ICW Workflow – File Protection
- ICW Workflow – Reopen ICW

Logging into Dell Unisphere for PowerMax

Steps

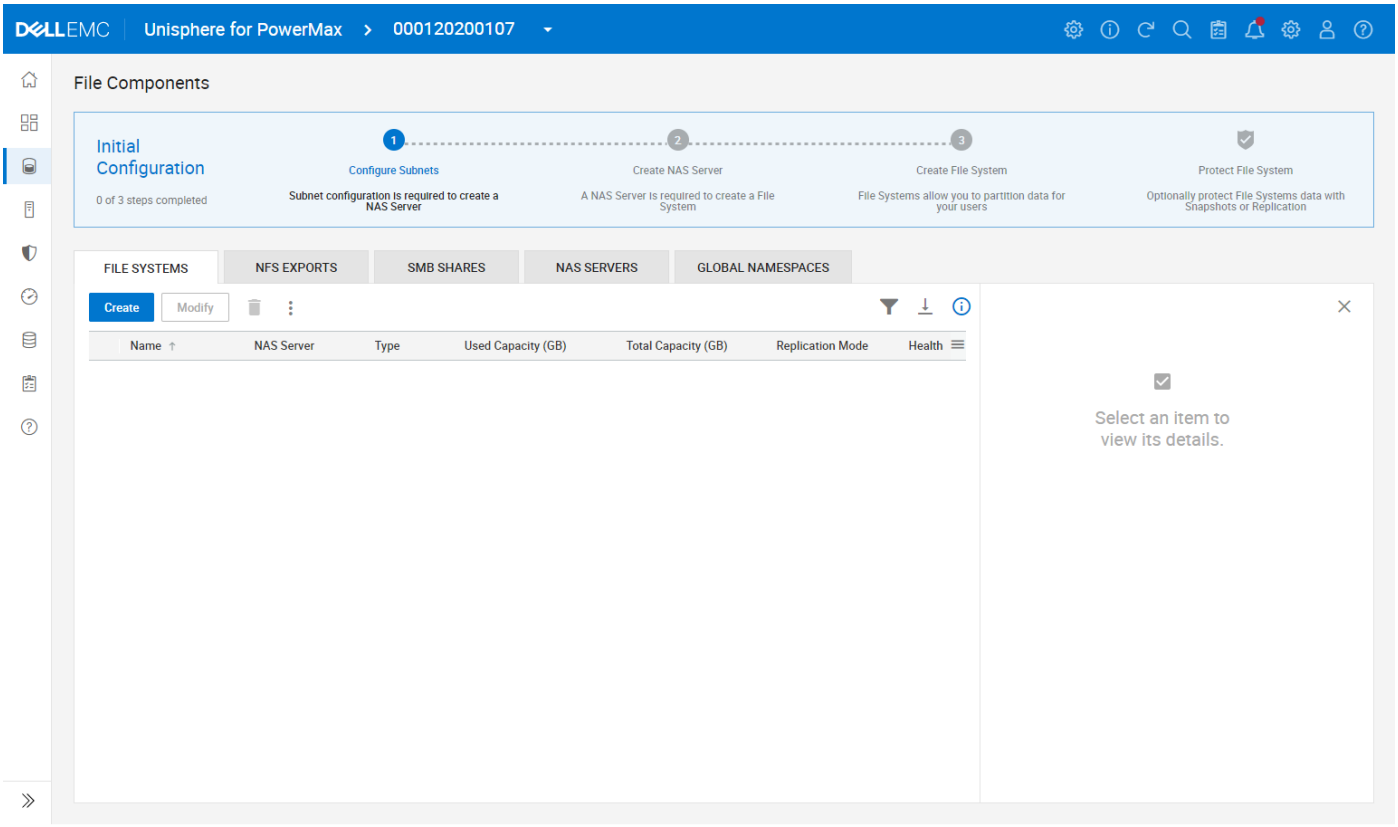
1. Use the following URL format to open Dell Unisphere for PowerMax: https://<unisphere_ip>:8443/
2. In the login dialog box, type the admin username and password. PowerMax File requires administrative privileges for File management.

Walk with Initial Configuration Wizard

The Initial Configuration Wizard (ICW) provides a guided tour of the File sections of Unisphere. It organizes the operations on File as a step-by-step procedure thereby simplifying the overall File configuration. The ICW facilitates setting up basic configuration and File protection spanning snapshot and replication. ICW can be closed after completion of the File configuration and reopened at any time to provide a handy link to command File areas.

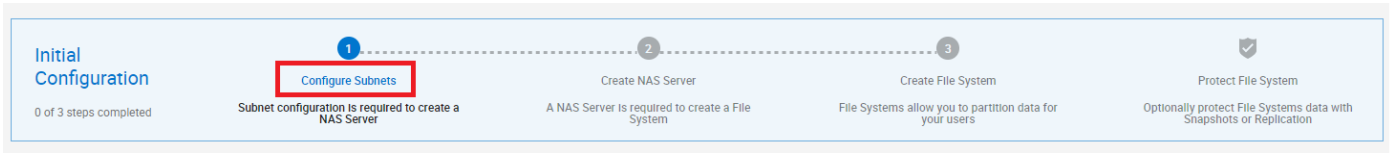
Launching the Initial Configuration Wizard (ICW)

ICW automatically appears on the first load of File. ICW can be viewed from any of the File pages (**Storage** > **File** > **Data Protection** > **File Protection** > **System** > **File Configuration**).

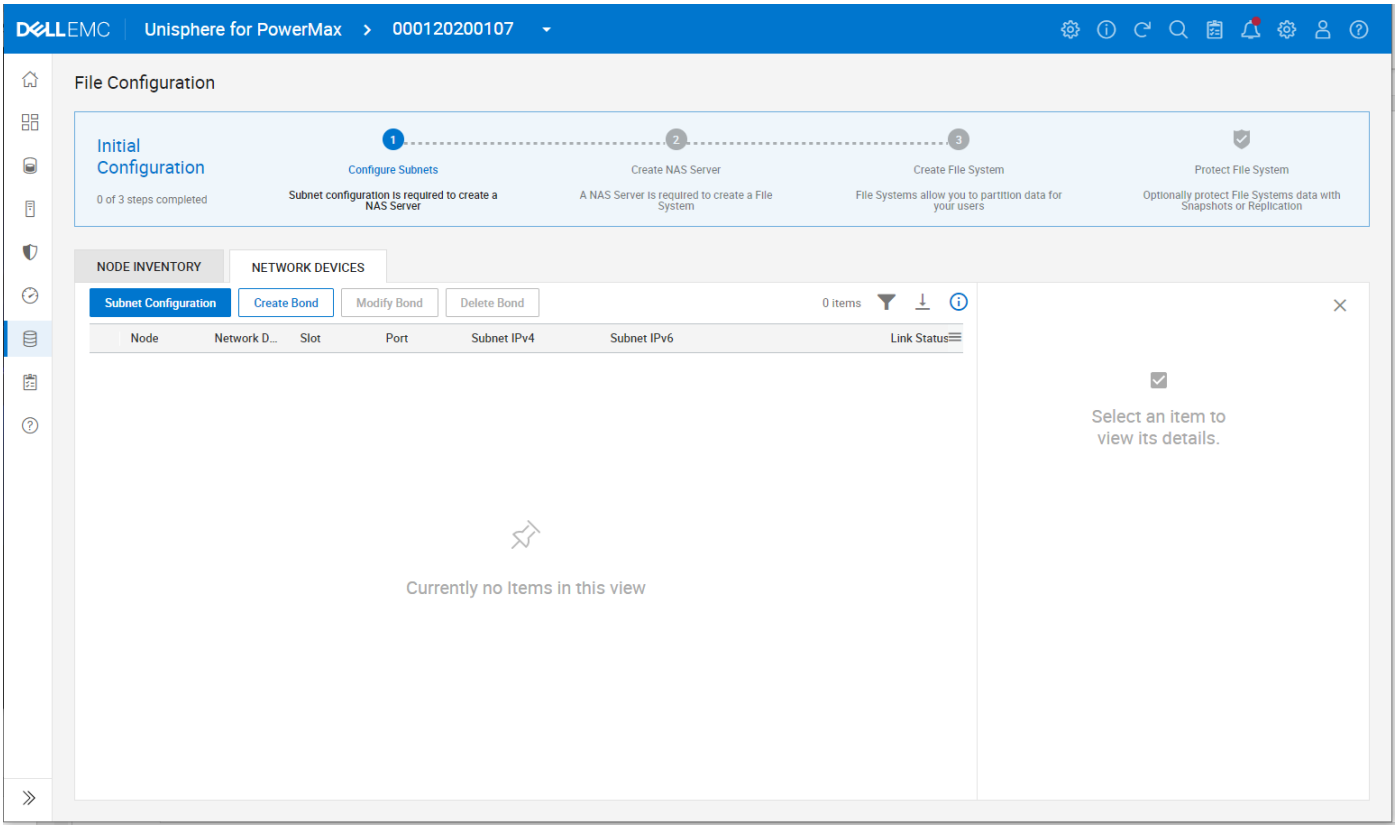


ICW Workflow – Configure Subnets

Each step title in the ICW provides a link to accomplish the workflow task. As a first step to setting up File, subnets need to be configured. Select **Configure Subnets**.

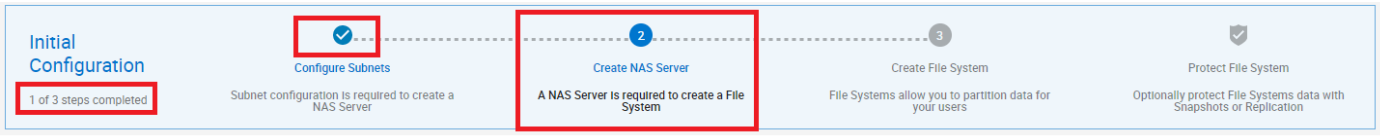


This will launch the **Network Devices** page. Selecting the **Subnet Configuration** button here will facilitate the setup of Subnets.

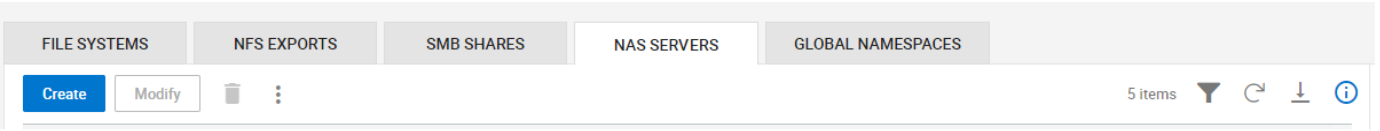


ICW Workflow – Create NAS Server

Once a step has been completed, the steps completed counter will update along with both the completed step and the next step. In this case, after the completion of **Configure Subnets**, the **Create NAS Server** step is activated.

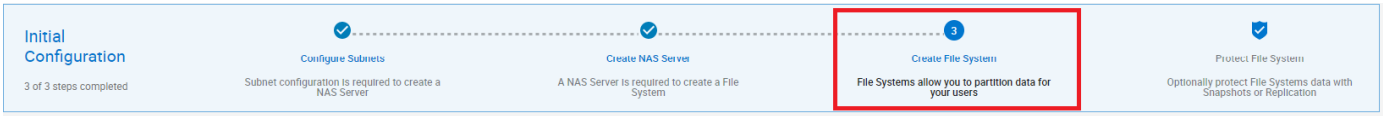


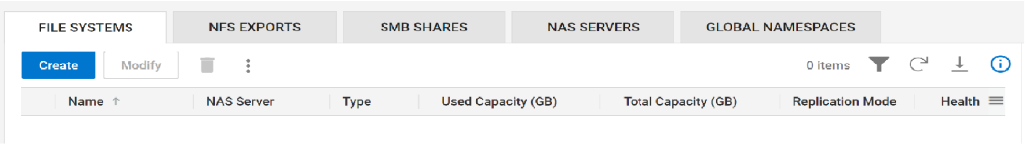
Selecting the **Create NAS Server** link will launch the **NAS Servers** tab. Selecting the **Create** button will allow creation of a NAS Server.



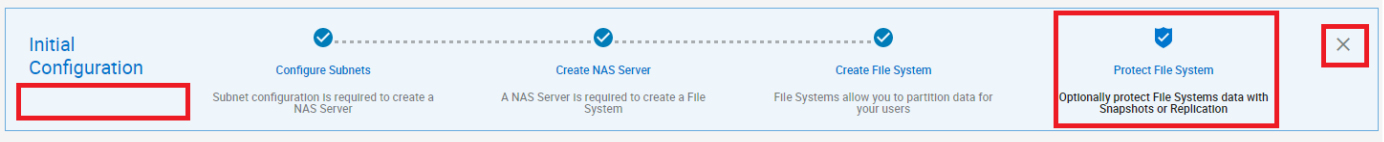
ICW Workflow – Create File System

Once the NAS Server is created, the **Create File System** step is activated. Selecting on this link will launch the **File Systems** tab from where new Filesystems can be created.



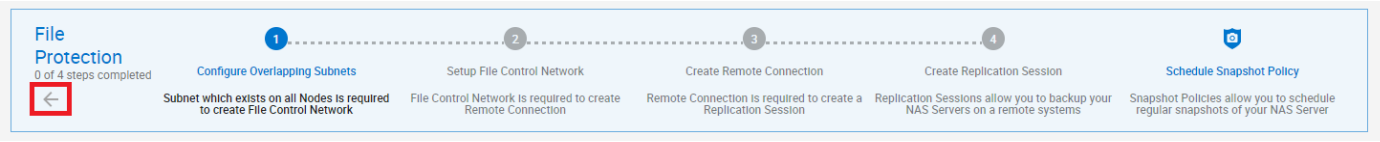


Once all steps have been completed, the **Protect File System** guided tour will become available and the steps completed counter will be removed. The ICW can now be closed.



ICW Workflow – File Protection

The File Protection guided tour work in a similar fashion to the ICW with only the inclusion of a back button to return to the ICW.



ICW Workflow – Reopen ICW

When closed the ICW can be reopened by clicking the information icon on the right-hand side of any of the File pages Titles.



3

Introduction to PowerMax File

This section provides a high-level overview of PowerMax File features.

File management and orchestrations are handled by Unisphere for PowerMax. All File features are exposed via both UI and REST API. The File alerts and performance data can be accessed from the common Alerts and Performance sections of Unisphere PowerMax. File is a highly available platform. For each File node, there exists a backup node. File also supports heterogeneous networking. File nodes need not have same SLIC type or number of SLICs and each node can have different subnets. However, the node denoted as 'Backup Node' for a NAS Server is expected to have the same subnets as its primary.

File fully automates its storage provisioning. For the given Filesystem and its size, the File platform automates the provisioning, masking of PowerMax volumes. File uses SnapVX for snapshots and SRDF/S or SRDF/A for synchronous or asynchronous replication support.

File supports 3-way NDMP and all back up software supporting NDMP can be used for backup. File also supports Dell VSI plug-in for VMware.

Following are some configuration limits:

Max FS Count (2 nodes)	2048
Max FS Count (more than 2 nodes)	4096
Max FS Size	64TB

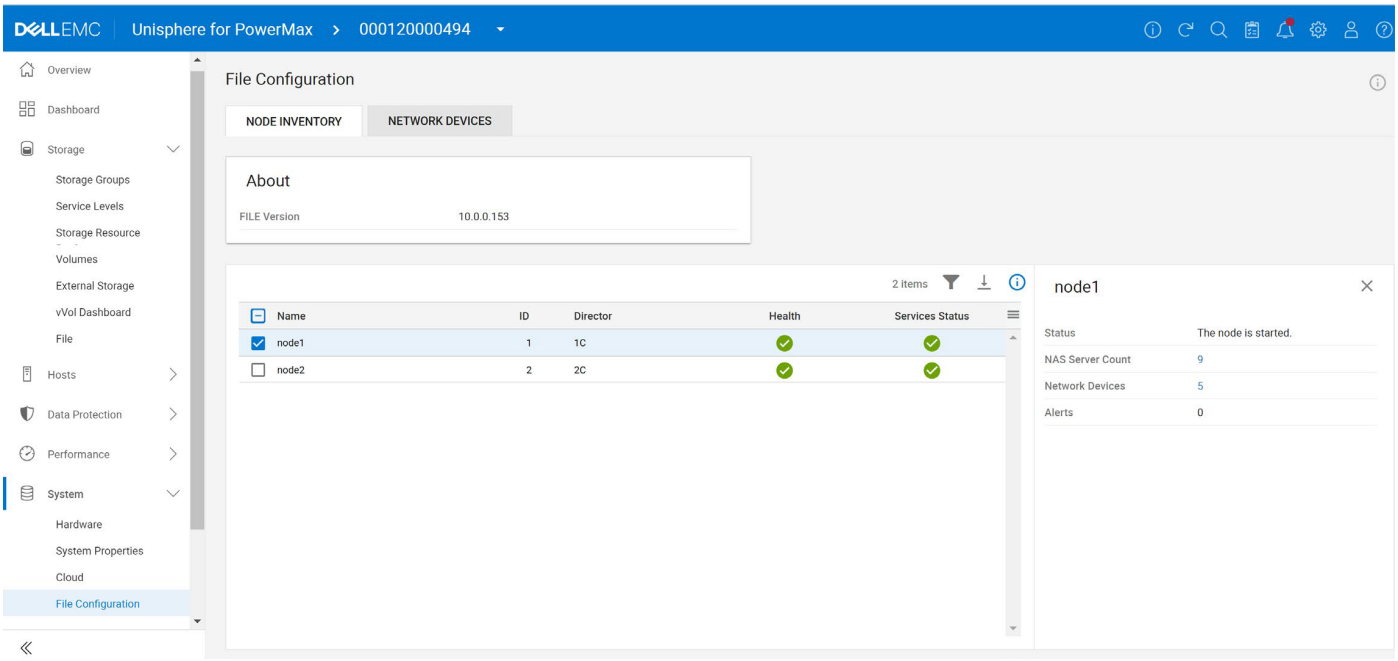
NOTE: The File system count includes checkpoint snapshots.

Topics:

- System Inventory
- Networking
- NAS Server

System Inventory

The **Node Inventory** page can be accessed from **System** > **File Configuration** in Unisphere for PowerMax. This page provides information of the File Version and number of nodes with their health status. Upon selection of a File node, the inventory provides configuration details on the number of NAS Servers, network devices configured for the File node. Any alert pertaining to the File Node can also be accessed from the **Alerts** list.



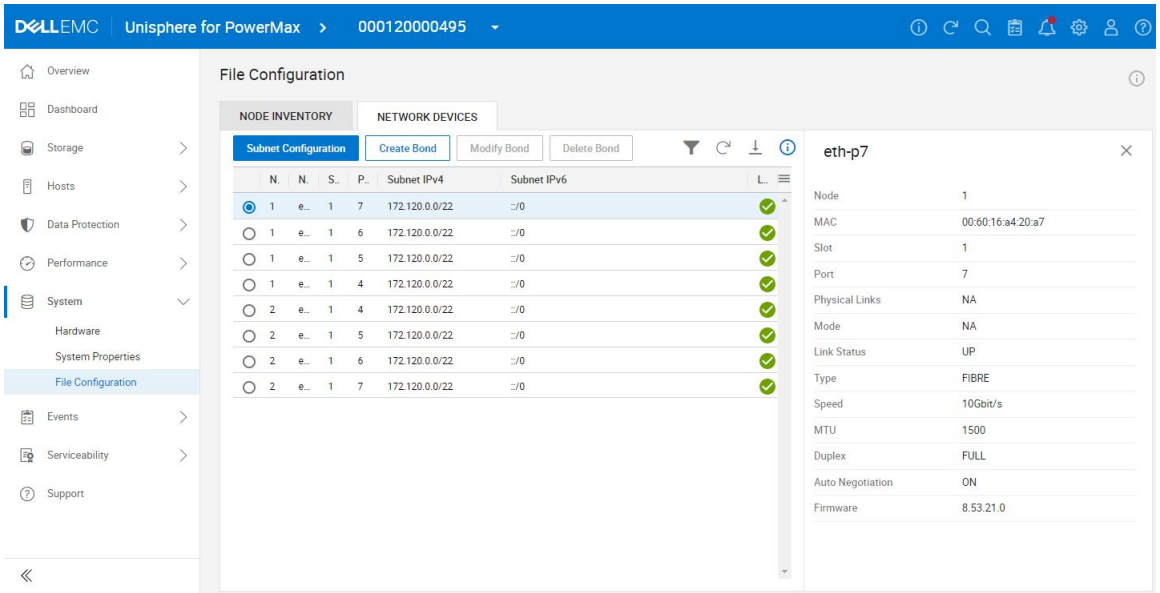
Networking

PowerMax File support 25GbE SLICs. File networking supports LACP and FSN protocols. PowerMax File supports heterogenous network configuration. Each File node can support varying number of SLICs and so to enable a seamless network failover and bonding support, it is required to provide all the existing Customer subnet configuration that are likely to participate in File network configuration.

NOTE: VLAN tagging and Jumbo frames are currently not supported for PowerMax File.

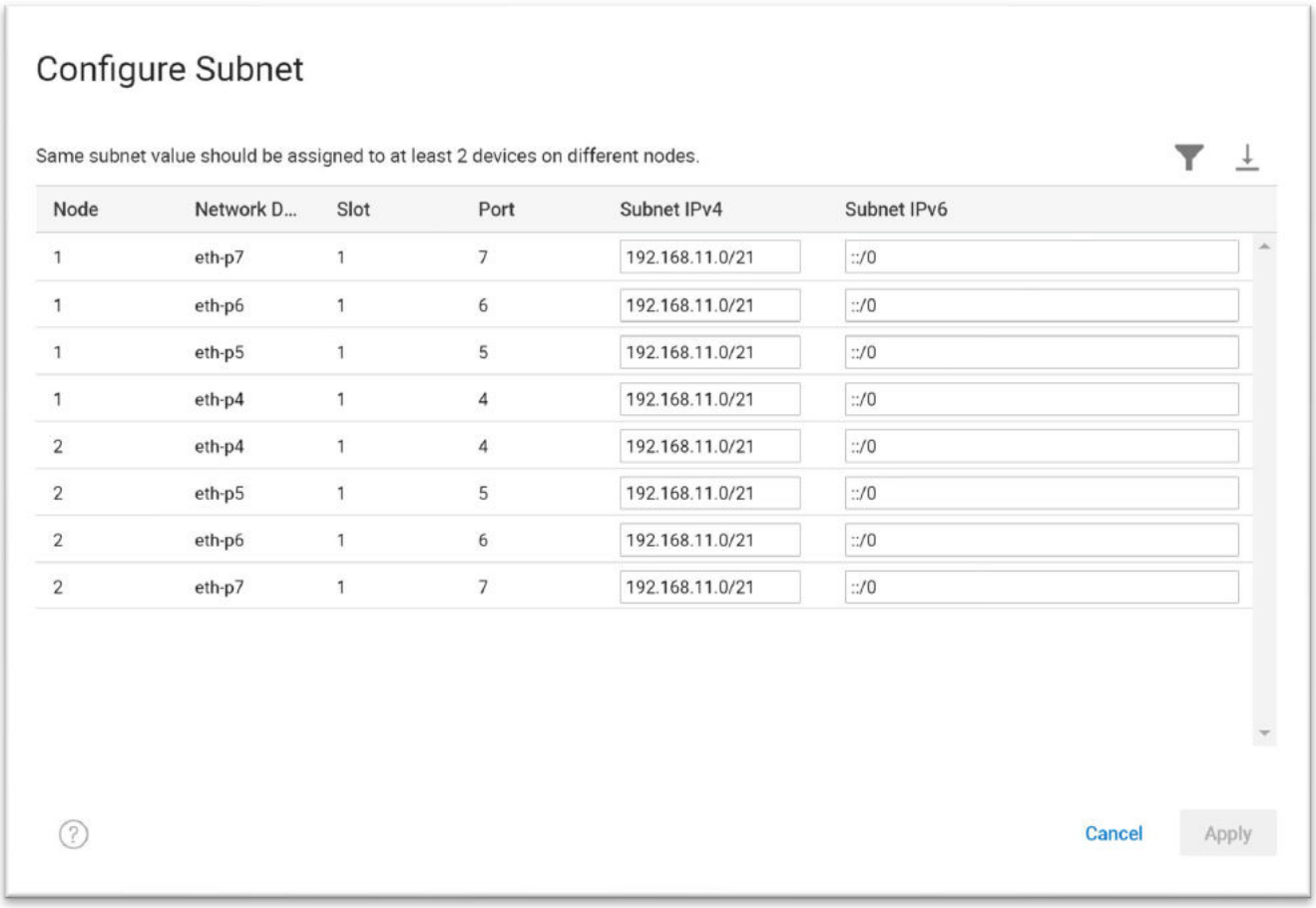
Listing Subnets

The **Network Devices** tab under the **System** > **File Configuration** lists all the subnets that are configured in a given user environment.



Assign Subnets

New subnets can be added by selecting the **Subnet Configuration** button. The configure subnet view allows adding new IPv4 and/or IPv6 subnets. By default, IPv4 subnets are set to 0.0.0.0/0 and IPv6 to ::/0. User can edit them with appropriate subnets. Note that a subnet must be configured on atleast 2 different ports on 2 different nodes. This is required for failover considerations. The **Configure Subnet** page can be used to modify the subnet as well.



If a single node is configured with the same subnet on 2 or more SLICs, they can be grouped to form a SLIC bond. There are 2 type of bonding – Active Backup or DLA (Dynamic Link Aggregation). In **Active Backup** mode, one SLIC is active while the other is passive and **DLA** supports active-active. Bonding can be created by selecting **Create Bond** button from **Network Devices** tab. SLIC bonding can be modified or deleted. Select the **Modify Bond** or **Delete Bond** for these operations.

Create Bond Device

Select Node *

Node:1, Director:1C

Select Mode *

Active-Backup

Select Network Devices (Minimum 2)

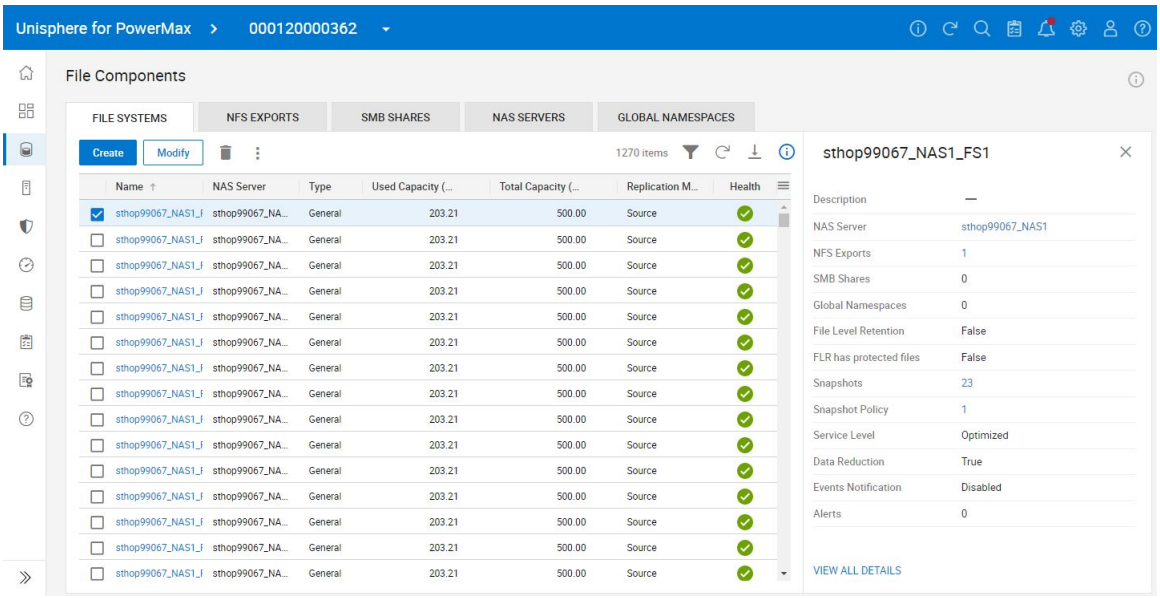
	Networ...	Slot	Port	Subnet IPv4	Subnet IPv6	Link Status
☐	eth-p5	1	5	172.20.50.1/21	::/0	✓
☐	eth-p4	1	4	172.20.50.1/21	::/0	✓

Cancel OK

NAS Server

Listing NAS Server

The NAS Server list view provides the list of NAS Servers configured on the system.



Create NAS Server

Prerequisites

You must create NAS servers before creating file systems.

Ensure to have your NAS network information available.

Steps

1. Select **Storage** > **File**.
2. Click the **NAS SERVERS** tab.
3. Select **Create**.
4. Continue to work through the **Create NAS Server** wizard.

Wizard Screen	Description
General Details	Enter a unique NAS Server name. Select a primary Node and Backup Node. **NOTE:** Primary and Backup node can not be the same. Select Storage Resource Pool.
File Interface	**Select Subnet** Select Network Device on Primary Node and Select Network Device on Backup Node. Enter IPv4/IPv6 Address and Gateway. **NOTE:** IP address and gateway should be from the usable host IP range in the selected subnet.

Wizard Screen	Description
Protocol	Select NFS and SMB server. **NOTE:** Configure either NFS or SMB or both protocols as per the requirement. **NOTE:** SMB server can be configured as a standalone server or joined to active directory. Enter a SMB Computer Name, SMB Server Description, Windows Domain and Domain Privileged Username.
DNS	Click the Enable DNS checkbox. **NOTE:** DNS server information is mandatory when: • Joining an AD domain, but optional for a stand-alone NAS server. • Configuring Secure NFS. Ener DNS Domain. Add or Delete IP address.
UDS	Select the checkbox Enable a Unix Directory service using NIS or LDAP. **NOTE:** UDS can be configured using Local Files, NIS or LDAP. If more than 1 service is configured user can set the order of their usage. Enter NIS domain Add or Delete IP address.
Summary	Review the content and Select **Back** to go back and make any corrections.

5. Select **Run In The Background** to create the NAS server.

   Once you have created the NAS server for NFS, you can continue to configure the server settings. If you enabled Secure NFS, you must continue to configure Kerberos.

   Select the NAS server to continue to configure or edit the NAS server settings.

Create File Interfaces

Prerequisites

Multiple File Interface can be created on a NasServer. Production network interfaces are used for all file protocols and services of NAS server. Backup network interfaces are used only for NDMP/NFS backup. Device on the primary/ backup node can be a virtual port or link aggregation on which the network interface is running. Gateway is an optional attribute.

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **NETWORK** > **FILE INTERFACE**.
2. Click **Create**.

   Create File Interface wizard opens.
3. Select **Subnet**.
4. Select Network Device On Primary Node.
5. Select Network Device On Backup Node.
6. Enter IP address, Gateway and select Role.
7. Click OK.

Change Preferred Interfaces

Prerequisites

The preferred interface of NAS Server is an interface from which all the non-local outbound connections of this NAS Server are initiated. The preferred interfaces for IPv4 and IPv6 are independent from each other. The acting preferred interfaces, one for IPv4 and one for IPv6, are selected among active interfaces by the following rules, ordered by priority (highest first):

- Manually selected interfaces have priority over automatically selected ones.
- Production interfaces have priority over backup and DR testing ones.
- The interface with the default gateway has priority over one not having one.
- From the otherwise equal priority interfaces, one with the most routes has the priority.
- From the otherwise equal priority interfaces, one with the minimal value of IP address has the priority.

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **NETWORK** > **CHANGE PREFERRED INTERFACE**.
2. Click **Change Preferred Interface**.
3. Select **Production IPv4 and IPv6**.
4. Select **Backup IPv4 and IPv6**.
5. Click **OK**.

Setup SMB Server

SMB, which stands for Server Message Block, is a protocol for sharing files and communications abstractions such as named pipes and mail slots between computers. SMB was also known as Common Internet File System (SMB).

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **SHARING PROTOCOLS** > **SMB Server**.
2. If Disabled is on, slide the button to change to **Enable**.
3. Select the radio button **Join to Active Directory Domain**.
4. Enter SMB Computer Name, SMB Server Description, and Windows Domain.

   NOTE:
   - In case a server that belongs to a Windows domain (Active Directory), DNS must be enabled on the NAS server.
   - The credentials of an administrator of the domain are required to join that domain.

5. Click **Apply**.

Setup NFS Server

Prerequisites

User can configure one NFS server per NAS Server. NFS servers support the NFSv3 and NFSv4 protocols.

NOTE: Secure NFS can be configured with following pre-conditions.

1. DNS and NTP must be configured
2. UDS must be configured using NIS, LDAP or Local Files
3. NasServer must have Kerberos configured or SMB joined to Active Directory

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **SHARING PROTOCOLS** > **NFS Server**.
2. If Disabled is on, slide the button to change to **Enable**.
3. Select checkbox NFSv3.

4. Click **Apply**.

Setup FTP Server

One FTP server and one SFTP server per NAS server can be configured. File Transfer Protocol (FTP) is a standard network protocol used to transfer files from one host to another host over a TCP-based network, such as the Internet.

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **SHARING PROTOCOLS** > **FTP**.
2. If Disabled is on, slide the button to change to **Enable FTP** and **Enable SFTP**.
3. Click **Apply**.

User Mapping

Prerequisites

NOTE:

- Default Unix User : Name used for granting access in case of Windows to Unix user mapping failure. When empty, access in such case is denied.
- Default Windows User: Name used for granting access in case of Unix to Windows user mapping failure. When empty, access in such case is denied.
- Auto User Mapping: A windows user must have a corresponding matching unix user (uid) in order to connect. This attribute allow to automatically generates this unix user (uid) if that windows user doesn't have any in the configured unix directory service (UDS). In a pure SMB or non-multi-protocol environment this should be set to true.
- UDS Search Order : Define the Unix directory service used for looking up identity information for Unix such as UIDs, GIDs, net groups, and so on:
  - 0 : None
  - 1 : NIS
  - 2 : LDAP
  - 3 : Local files
  - 4 : Local then NIS
  - 5 : Local then LDAP

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **SHARING PROTOCOLS** > **USER MAPPING**.
2. Select the **Unix Directory Service Search Order**.
3. Click **Apply**.

Setup Naming Services

DNS

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **NAMING SERVICES** > **DNS**.
2. If Disabled is on, slide the button to change to **Enabled**.
3. Enter **DNS Domain**.
4. Enter **IP Address** and click **Add**.

5. Click **Apply**.

UDS – NIS

The Network Information Service (NIS) consists of a directory service protocol for maintaining and distributing system configuration information, such as user and group information, hostnames and so on.

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **NAMING SERVICES** > **UDS**.
2. If Disabled is on, slide the button to change to **Enabled**.
3. In the Unix Directory Service drop down list select **NIS**.
4. Enter **NIS Domain**.
5. Enter **IP Address** and click **Add**.
6. Click **Apply**.

UDS – LDAP

The Lightweight Directory Access Protocol (LDAP) is an application protocol for querying and modifying directory services running on TCP/IP networks.

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **NAMING SERVICES** > **UDS**.
2. If Disabled is on, slide the button to change to **Enabled**.
3. In the Unix Directory Service drop down list select **LDAP**.
4. Enter LDAP Domain.
5. Enter **IP Address** and click **Add**.
6. Click **Apply**.

Local Files

Prerequisites

NOTE: Supported files are:

- Passed File
- Hosts File
- Group File
- Netgroup File
- Nsswitch.conf File
- Homedir File
- Ntxmap File

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **NAMING SERVICES** > **LOCAL FILES**.
2. If Disabled is on, slide the button to change to **Enabled**.
3. Click **Apply**.

Setup Antivirus

The storage system supports third-party anti-virus servers that perform virus scans and reports back to the storage system.

Prerequisites

NOTE:

- During the scan any access to this file is blocked.
- If the file does not contain a virus, it is written to the file system. If the file is infected, corrective action (fixed, removed or placed in quarantine) is taken as defined by the anti-virus server.
- User can optionally set up the service to scan the file on read access based on last access of the file compared to last update of the third-party anti-virus date.
- A configuration file (named viruschecker.conf) needs to be uploaded before enabling the anti-virus service.

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **SECURITY** > **ANTIVIRUS**.
2. If Disabled is on, slide the button to change to **Enabled**.
3. Begin the Antivirus Scanning by sliding the button to change to **Start**.
4. Click **Apply**.

Set up Kerberos

Kerberos is a distributed authentication service designed to provide strong authentication with secret-key cryptography.

Prerequisites

NOTE:

- It works based on `tickets` that allow nodes communicating over a non-secure network to prove their identity in a secure manner.
- When configured to act as a secure NFS server, the Nas Server uses the RPCSEC_GSS security framework and Kerberos authentication protocol to verify users and services.
- User can configure a secure NFS environment for a multiprotocol NAS Server or one that supports Unix-only shares. In this environment, user access to NFS file systems is granted based on Kerberos principal names.
- Up to 10 Kerberos Key Distribution Center (KDC) servers can be added.

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **SECURITY** > **KERBEROS**.
2. If Disabled is on, slide the button to change to **Enabled**.
3. Enter **Realm**.
4. Add **IP Address**.
5. Enter **TCP Port**.
6. Upload KeyTab File or Retrieve KeyTab File

   NOTE: Kerberos must be configured before uploading KeyTab File. The KeyTab is required only for secure NFS.

7. Click **Apply**.

Setup NDMP

The Network Data Management Protocol (NDMP) provides a standard for backing up file servers on a network.

Prerequisites

NOTE:

- File supports 3-way NDMP only.
- NDMP allows centralized applications to back up file servers that run on various platforms and platform versions.
- NDMP reduces network congestion by isolating control path traffic from data path traffic, which permits centrally managed and monitored local backup operations.
- Storage systems support NDMP v2-v4 over the network.
- Direct-attach NDMP is not supported. This means that the tape drives need to be connected to a media server, and the NAS server communicates with the media server over the network.
- NDMP has an advantage when using multiprotocol file systems because it backs up the Windows ACLs as well as the UNIX security information.

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **BACKUP & EVENTS** > **NDMP**.
2. If Disabled is on, slide the button to change to **Enabled**.
3. Enter **Username**.
4. Enter **Password**.
5. Enter **Confirm Password**.
6. Click **Apply**.

DHSM

DHSM (Distributed Hierarchical Storage Management) automates the archiving of NAS files across a hierarchy of storage platforms.

Prerequisites

NOTE:

- When used with an external policy and archiving application such as Glider, DHSM automatically archives infrequently used files, or files of a particular type, to less-expensive storage devices.
- Archived files are then recalled to NAS storage when accessed by client applications.

About this task

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **BACKUP & EVENTS** > **DHSM**.
2. If Disabled is on, slide the button to change to **Enabled**.
3. Enter **Username**.
4. Enter **Password**.
5. Enter **Confirm Password**.
6. Click **Apply**.

Results

Example

Events Pool Publisher

File Event Service pool is a pool of remote File Event Service servers (machines that run VEE and are capable to handle event notifications from the NAS Server).

Prerequisites

NOTE:

- NAS Server can have one or several (up to three) File Event Service pools.
- The File Event Service pool servers are responsible for:
  - Maintaining a topology and state mapping of all consumer applications
  - Delivering event type and associated event metadata through the publishing agent API

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **BACKUP & EVENTS** > **NDMP**.
2. Click **Create**.
   Create Events Publishing Pool wizard opens.
3. Enter **Pool Name**.
4. Add **Events Publishing Server**.
5. Under PRE-EVENTS select the checkbox **Select All**.
6. Click **OK**.

Events Publisher

File Event Service is a mechanism whereby applications can register to receive events notifications and context from sources such as SDNAS.

Prerequisites

NOTE:

- File Event Service is a part of Common Event Enabler (CEE).
- CEE provides the working environment for the CAVA (Common Antivirus Agent) and CEPA (Common Event Publishing Agent) facilities.
  - The Common Event Publishing Agent delivers to the application both event notification and associated context in one message.
  - Context may consist in file metadata or directory metadata needed to decide business policy.

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **BACKUP & EVENTS** > **EVENTS POOL**.
2. Click **Create**.
   Create Events Publishing Pool wizard opens.
3. If Disabled is on, slide the button to change to **Enabled**.
4. Select the **Post Event Failure Policy** from the drop-down list.
5. Select **Using SMB Server Account**.
6. Enter **Heartbeat**.
7. Enter **Timeout**.
8. Click **OK**.

Node

A File node is a single entity container, member of an SDNAS cluster. Each node communicates with each other and work toward a common goal.

Primary node:

- One of the SDNAS cluster nodes, on which the NAS server is currently running.
- The node is chosen during NAS server creation according to user supplied input.

Backup Node:

- One of the SDNAS cluster nodes, on which the NAS server is backed up for fault tolerance purposes.
- This means that the NAS server will be moved to this node during any failover event.
- The node is chosen during NAS server creation according to user supplied input.

Node – Change Primary

Prerequisites

NOTE:

- As a result of this action the NAS Server will automatically be moved to its new Primary Node.
- Provided Primary Node value can not be equal to the NAS Server Backup Node - the request will be rejected.
- For seamless movement of the File Interfaces, user needs to select the devices on newly provided node corresponding to every interface.

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **NODES** > **Change Primary Node**.
2. In the Change Primary Note wizard, Enter **Current Primary Node**.
3. Select the **Primary Node**.
4. Select Network device for each of the file interfaces in the New Primary node.
5. Click **OK**.

Node – Change Primary

Prerequisites

NOTE:

- As a result of this action the provided Node will be assigned with the Backup role for the specified NAS Server. There will not be any immediate movement of NAS Server.
- The previous Backup Node will no longer serve as a Backup for specified NAS Server.
- Provided Backup Node value can not be equal to the NAS Server Primary Node - the request will be rejected.
- For seamless movement of the File Interfaces, user needs to select the devices on newly provided node corresponding to every interface.

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **NODES** > **Change Backup Node**.
2. In the Change Primary Note wizard, Enter **Current Backup Node**.
3. Select the **Backup Node**.
4. Select Network device for each of the file interfaces in the New Primary node.
5. Click **OK**.

Node – Swap Nodes

Prerequisites

NOTE:

- Use this command if you'd like to change the current Primary Node and move the NAS Server to its current Backup Node.
- As a result of this action the current Primary Node will become Backup Node and current Backup Node will become Primary Node.
- This action can be used after failover of NAS Server due to node failure. It will bring the NAS Server to the node that it was hosted on prior the failover.

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **NODES** > **Swap Nodes**.
2. A **Warning** dialogue box opens.
3. Click **OK**.

Filesystem

PowerMax File leverages a mature and robust 64-bit filesystem that is designed for scalability, efficiency, flexibility and high performance. It supports both traditional and transactional NAS use cases where transactional NAS is limited to VMWare support for Juniper. Approximately 1.5GB is reserved for metadata. For example, a 100GB filesystem will have a usable capacity of 98.5GB. PowerMax File also supports block-level compression, dedupe and encryption.

List Filesystem

About this task

To view the list of File Systems:

Steps

1. Select **Storage** > **File** > **NAS Servers**. The list of NAS server opens.
2. Click **VIEW ALL DETAILS** to access FS configuration.
   The following information can be viewed in the File System List view:
   - Type: General or Transactional
   - Source/Destination for replicated FS
   - Health of filesystem
   - Number of exports
   - Number of Snapshots
   - Filtered alerts if health is no OK

Create Traditional Filesystem

Steps

1. Select **Storage** > **File** > **File Systems**.
2. Click **Create**. Create File System wizard opens.
3. In Type tab, select General Filesystem and click **Next**.
4. In Select a NAS Server tab, select a NAS server to host the filesystem and click Next.

5. In File System Details tab, enter **FS name**, enter the size, under Storage Resource Pool select **SRP** if one or more exist, under Service Level set the block SLO, Data Reduction checkbox is selected by default, set optional threshold warning, under Advanced setting select **Oplocks Enabled** if NAS server supports SMB and click **Next**.
6. In FLR tab, set the FLR (Write Once Read Many – WORM FS) and click **Next**.

   NOTE: This option cannot be modified once it is created.

7. In NFS Export tab, enable the **Configure the initial Export**, enter share name, and click **Next**.
8. In Configure Access tab, enter Minimum security level (supports Kerberos), select Default Access, enable host-based access, and click **Next**.
9. In SMB Share tab, enable the **Configure the Initial Share**, provide SMB share name, and click **Next**.
10. In Summary tab, review the summary. Go back to make any necessary updates and click Run In The Background to create the FS.

Create Traditional Filesystem

Steps

1. Select **Storage** > **File** > **File Systems**.
2. Click **Create**. Create File System wizard opens.
3. In Select Type tab, select VMWare Filesystem and click **Next**.

   NOTE: transactional NAS is supported for VMWare only.

4. In Select a NAS Server tab, select a NAS server and click **Next**.
5. In File System Details tab, enter **name**, enter the **size**, select Service Level, Data Reduction checkbox is selected by default, under Advanced setting select the **block size**, and click **Next**.
6. In NFS Export tab, enter the Name, and click **Next**.
7. In Configure Access tab, enter Minimum security level (supports Kerberos, host-based access), and click **Next**.
8. In Summary tab, review the summary. Go back to make any necessary updates and click Run In The Background.

Modify Filesystem

Modification for FLR filesystems is allowed for the following cases:

- Retention periods
- Enable/Disable auto-lock of new files
- Set auto-lock policy interval
- Enable/Disable automatic deletion of files on retention period expiration.

NOTE: FLR cannot be set after a Filesystem is created. It must be part of the Filesystem creation process.

Modify Traditional Filesystem

Steps

1. Select **Storage** > **File** > **File Systems** > **[nas server]** > **FILELEVEL RETENTION**.
2. Click **Modify**. Modify File System wizard opens.
3. Enter Description, select Size, select Service Level, under Advanced Settings select **Oplocks Enabled**, select **Access Policy**, select **Folder Rename Policy**, select **Locking Policy**.

   NOTE: Traditional Filesystem supports SMB property updates, if configured.

4. Click **OK**.

Modify Transactional Filesystem

Steps

1. Select **Storage** > **File** > **File Systems** > **[nas server]** > **FILELEVEL RETENTION**.
2. Click **Modify**. Modify File System wizard opens.
3. Enter Description, select Size, select **Service Level**.
4. Click **OK**.

Delete File System

Prerequisites

This option supports deletion of Traditional and Transactional Filesystem. FLR filesystems can be deleted if no protected files exist.

Steps

1. Select **Storage** > **File** > **File Systems** > > **select the checkbox against the [nas server]**.
2. Click **Delete icon**. Delete File System dialogue box opens.
3. Click **OK**.

Setup Filesystem Quotas

PowerMax File supports User & Tree quotas. User Quotas can also be supported over Tree quotas. Quotas for Filesystem can be managed from Unisphere for PowerMax or from Windows Clients (for SMB-based Filesystem). It supports hard & soft quota limits. Administrators can create/modify/delete quotas. Quotas can also be accessed from 'All details' view of the selected filesystem.

User Quotas

Steps

1. Select **Storage** > **File** > **File Systems** > **[nas server]** > **QUOTAS** > **USER QUOTAS**.
2. Click **Create**.
3. Enter Windows SID
4. Click **OK**.

Tree Quotas

Steps

1. Select **Storage** > **File** > **File Systems** > **[nas server]** > **QUOTAS** > **TREE QUOTAS**.
2. Select **Add**.
3. Slide the **Enforce User Quota**to the right to enabled User Quota defaults on the Tree Quota.
4. Enter a **Grace Period** to count down the time between the soft and hard limit. You will begin to receive alerts once the grace period is reached. To track space consumption without setting limits, set the **Soft Limit** and **Hard Limit** fields to 0, which indicates no limit.
5. Click **OK**.

Protocols

Sharing Protocols – SMB Server

SMB, which stands for Server Message Block, is a protocol for sharing files and communications abstractions such as named pipes and mail slots between computers. SMB was also known as Common Internet File System (SMB).

Steps

1. Select the **Storage** > **File** > **NAS Servers** > **[nas server]** > **Sharing Protocols** > **SMB Server** tab.
2. If **Disabled** is on, slide the button to change to **Enabled** to define the NAS server.
3. Select **Join to the Active Directory Domain**.
4. Enter **SMB Computer Name**.
5. Enter **SMB Server Description**.
6. Enter **Windows Domain**.
7. Click **Apply**.

Sharing Protocols – NFS Server

Prerequisites

User can configure one NFS server per NAS Server. NFS servers support the NFSv3 and NFSv4 protocols.

NOTE: Secure NFS can be configured with following pre-conditions.

- DNS and NTP must be configured.
- UDS must be configured using NIS, LDAP or Local Files.
- NasServer must have Kerberos configured or SMB joined to Active Directory.

Steps

1. Select the **Storage** > **File** > **NAS Servers** > **[nas server]** > **Sharing Protocols** > **NFS Server** tab.
2. If **Disabled** is on, slide the button to change to **Enabled** to define the NAS server.
3. Enable either NFSv3, NFSv4, or both.
4. Click **Apply**.

Sharing Protocols – FTP Server

Prerequisites

You can configure one FTP server and one SFTP server per NAS server. File Transfer Protocol (FTP) is a standard network protocol used to transfer files from one host to another host over a TCP-based network, such as the Internet.

NOTE:

- The FTP and SFTP clients are authenticated using credentials defined on a Unix name server (such as a NIS server, an LDAP server or a local passwd file) or a Windows domain.
- FTP client may connect as anonymous user. This requires a Unix user named 'ftp' in the name server.

Steps

1. Select the **Storage** > **File** > **NAS Servers** > **[nas server]** > **Sharing Protocols** > **FTP** tab.
2. Under FTP, if **Disabled** is on, slide the button to change to **Enable**.
3. Optionally also enable SSH FTP. Under **SFTP**, if Disabled in on, slide the button to **Enable**.
4. Select which type of authenticated users have access to the files.

5. Optionally, show the **Home Directory and Audit** options.
6. Select or clear the **Home directory restrictions**. If disabled, enter the **Default home directory**.
7. Select or clear **Enable FTP/SFTP Auditing**. If checked, enter the directory location of where to save the audit files, and the maximum size allowed for the audit file.
   - Optionally, **Show Messages**, and enter a default welcome message, and message of the day.
   - Optionally, show the **Access Control List**, and add a list of users, groups, and hosts that are allowed, or denied FTP access.
8. Click **Apply**.

Export and SMB View

Create NFS exports

Each NFS share must have a unique local path. Unisphere automatically assigns this path to the initial share created within a new file system. The local path name is based on the file system name. Before creating additional shares within an NFS file system, create a directory to share from a Linux/UNIX host which is connected to the file system. Then, create a share from Unisphere and set access permissions accordingly.

Prerequisites – A NAS server must be configured to support NFS protocol before creating a file system.

Export View

Steps

Select **Storage** > **File** > **NFS Exports**.

The list of NFS Exports opens.

Create

Steps

1. Select **Storage** > **File** > **NFS Exports**.
2. Click **Create**. Create NFS Export wizard opens.
3. In **Select File System** tab, select a file system to create your NFS Export ad click Next.
4. In Select Snapshot (Optional) tab, select a snapshot and click Next.
5. In Create NFS Export Details tab, enter Name, description, and Local Path. Click Next.
6. In Configure Access tab, select minimum security, default access, click Create and click Next.
7. In Summary tab, review the content and click **Run In The Background**.

Modify

Steps

1. Select **Storage** > **File** > **NFS Exports**.
2. Click **Modify**. Modify NFS Export wizard opens.
3. Enter **Description**.
4. Click **OK**.

Delete

Steps

1. Select **Storage** > **File** > **NFS Exports**.

2. Click **Delete** icon. Delete NFS Export dialogue box opens.
3. Click **OK**.

Host Access

Steps

1. Select **Storage** > **File** > **NFS Exports**.
2. Click **Host Access**. Host Access wizard opens.
3. Select Minimum Security and Default Access.
4. Click Create to add host.
5. Click Import to import host via csv file.
6. Click **OK**.

Create NFS exports

An SMB file system allows you to create multiple shares with the same local path. In these cases, you can specify different host-side access controls for different users, but the shares within the file system will all access common content.

A directory must exist before you can create shares on it. Therefore, if you want the SMB shares within the same file system to access different content, you must first create a directory on the Windows host that is mapped to the file system. Then, you can create corresponding shares using Unisphere.

Prerequisites – A NAS server must be configured to support SMB protocol before creating a file system.

SMB Shares List View

Steps

Select **Storage** > **File** > **SMB SHARES**.

The list of SMB Shares opens.

Create SMB Share

Steps

1. Select **Storage** > **File** > **SMB SHARES**.
2. Click Create.
3. In **Select File System** tab, select a file system to create your NFS Export ad click Next.
4. In Select Snapshot (Optional) tab, select a snapshot and click Next.
5. In SMB Share Details tab, enter Name, description, and Local Path. Click Next.
6. In Advance SMB Settings tab, select offline availability, enter unmask and click Next.
7. In Summary tab, review the content and click **Run In The Background**.

Modify SMB Shares

Steps

1. Select **Storage** > **File** > **SMB Shares**.
2. Click **Modify**. Modify SMB Shares wizard opens.
3. Enter **Description**, Unmask, and select offline availability.
4. Click **OK**.

Delete SMB Shares

Steps

1. Select **Storage** > **File** > **SMB Shares**.
2. Click **Delete** icon. Delete SMB Shares dialogue box opens.
3. Click **OK**.

Snapshot, Snap Policies

List Snapshots

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **SNAPSHOTS**.
2. The list of snapshots can be viewed with the following details:
   - Access Type - Protocol/Snapshot
   - Creator Type
   - Creation Time
   - Expiration Time – Time to go live for the Snapshot
   - Health
   - The right pane gives details of weather Snap is mounted or not mounted, SLO and Compression.

Create Filesystem Snapshots

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **SNAPSHOTS**.
2. Click **Create**.
   The Create Snapshot of File System wizard opens.
3. Enter Name, description, select Service Level, select Local Retention Policy, and select File Snapshot Access Type.
4. Click **Run In The Background**.

Modify Snapshots

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **SNAPSHOTS**.
2. Click **Modify.**.
   The Create Snapshot of File System wizard opens.
3. Enter Name, description, select Service Level, select Local Retention Policy, and select File Snapshot Access Type.
4. Click **Run In The Background**.

Clone Snapshots

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **SNAPSHOTS**.
2. Click **More Actions** icon.

3. Click **Clone**.
   The Create Snapshot of File System wizard opens.
4. Enter Name, description, and select Service Level.
5. Click **Run In The Background**.

Restore Snapshots

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **SNAPSHOTS**.
2. Click **More Actions** icon.
3. Click **Restore**.
   The Restore File System from Snapshot wizard opens.
4. Enter Backup Snapshot Name and select Rename to File System.
5. Click **Run In The Background**.

Refresh Snapshots

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **SNAPSHOTS**.
2. Click **More Actions** icon.
3. Click **Refresh**.
   The Refresh Snapshot from File System dialogue box opens.
4. Click **Run In The Background**.

Delete Snapshots

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **SNAPSHOTS**.
2. Click **Delete** icon.
   The Delete File Snapshot dialogue box opens.
3. Click **Run In The Background**.

Snap Schedules

A snapshot schedule manages the life cycle of file system snapshots. A schedule is created at the NAS server level and is assigned to one or more file systems within the NAS server. A schedule follows a rule that decides when a snapshot is taken place and when a snapshot is expired based on retention policy. As shown in Figure 1, Schedule rules are persisted in nameDB on cluster VDM. A schedule is persisted in nameDB on the user VDM. By default, a schedule is replicated and stored within configFS on user VDM. For the cases of local snapshot, a schedule can be configured as non-replicated and is stored within rootFS on user VDM.

Our current design has two flavors of setting up a schedule: (1) a schedule follows a rule template, and the attributes of the template are set up from the REST API. We discuss details of this scheme in 4.2. (2) a schedule follows a policy/rule that is set up by the platform, for example, a Trident system, which will have a unified interface of data protection for both block and file system use cases. The details of the integration design are explored in 4.3.

Snapshot policy view

Steps

1. Select **Data Protection** > **File Protection** > **SNAPSHOT POLICY**.
2. The list of snapshot policy can be viewed with the following details:
   - NAS Server
   - File System - Number of File Systems this policy is attached to
   - Sync Replicated – Weather the Snapshot Policy must be replicated
   - The right pane gives details of when will snapshot be taken and of which type, and if Snapshot will be automatically deleted after the retention time has passed.

Create Snapshot policy

Steps

1. Select **Data Protection** > **File Protection** > **SNAPSHOT POLICY**.
2. Click **Create**.
3. Enter Nas Server, Name, Recovery Point Objective, and select File Snapshot Access Type.
4. Click **OK**.

NOTE: PowerMax File supports schedule-based snapshots for both protocol and checkpoint on all the Filesystems. Administrators can run up to 50 schedules concurrently in one hour. It is recommended to spread out snap schedules such that at any point of time there are no more than 50 snap create or snap expiry task executions. It is also recommended to not schedule snaps below hourly unit.

Modify Snapshot Policy

Steps

1. Select **Data Protection** > **File Protection** > **SNAPSHOT POLICY**.
2. Click **Modify**. Modify Snapshot Policy wizard opens.
3. Enter NAS server, Name, Recovery Point Objective and select File Snapshot Access Type.
4. Click **Run In The Background**.

Copy Snapshot Policy

Steps

1. Select **Data Protection** > **File Protection** > **SNAPSHOT POLICY**.
2. Click **Copy**. Modify Snapshot Policy wizard opens.
3. Select NAS Server and enter Name.
4. Click **Run In The Background**.

Attach Snapshot Policy

Steps

1. Select **Data Protection** > **File Protection** > **SNAPSHOT POLICY**.
2. Select a **[nas server]** by clicking on the checkbox.
3. Click **More Actions** icon.

4. Click **Attach Snapshot Policies**. Attach Snapshot Policies dialogue box opens.
5. Select the Snapshot Policies to be attached to the file system.
6. Click **Add**.

Replication

Remote replication is a feature by which each NAS Server is replicated to remote site/sites to provide data availability in case of Active site disaster.

NAS Replication Components / Actions

- File Control Network
  - Create
  - Modify
  - Delete
- Remote Connection (with file capabilities)
- Create Replication session
- Delete Replication session
- FS Add to a Protected NAS Server
- FS Del from a Protected NAS Server
- Pause/Stop Replication session
- Resume/Start Replication session
- Switchover/ Planned Failover of Replication session
- Unplanned Failover of Replication session
- Planned Failover w/ auto start of Replication session
- Disaster Recovery

For more information refer the *Dell EMC Replication Guide* for File.

Events and Alerts

File alerts have been integrated into U4P. File alerts will be visible on the alert list view. The type will be shown as FILE for file alerts. All operations available for block alerts is now also available for file alerts:

- Acknowledgement and deletion of alerts
- Viewing, enabling, disabling alert policies
- Setting up notifications

Accessing the alerts

Alerts List View

Steps

1. Click **Events** > **Alerts**.
2. The list of Alerts can be viewed.

Viewing Alert Policies

Steps

1. Select **Settings** icon from the top right toolbar. The Settings dialogue box appears.
2. Click **Alerts** > **Alert Policies**. The Alert Policies can now be viewed.

Enabling Alert Policies

Steps

1. Select **Settings** icon from the top right toolbar. The Settings dialogue box appears.
2. Click **Alerts** > **Alert Policies**.
3. Select the check box to enable any alert policy(s).
4. Click **Apply**.

Disabling Alert Policies

Steps

1. Select **Settings** icon from the top right toolbar. The Settings dialogue box appears.
2. Click **Alerts** > **Alert Policies**.
3. Unselect the check box to enable any alert policy(s).
4. Click **Apply**.

Setting up Notifications

Steps

1. Select **Settings** icon from the top right toolbar. The Settings dialogue box appears.
2. Click **Alerts** > **Alert Policies**.
3. Slide the button to configure Email and SNMP and enter all the required details.
4. Select the icons under **Alert Severity** to enable notification for alerts.
5. Click **Apply**.

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